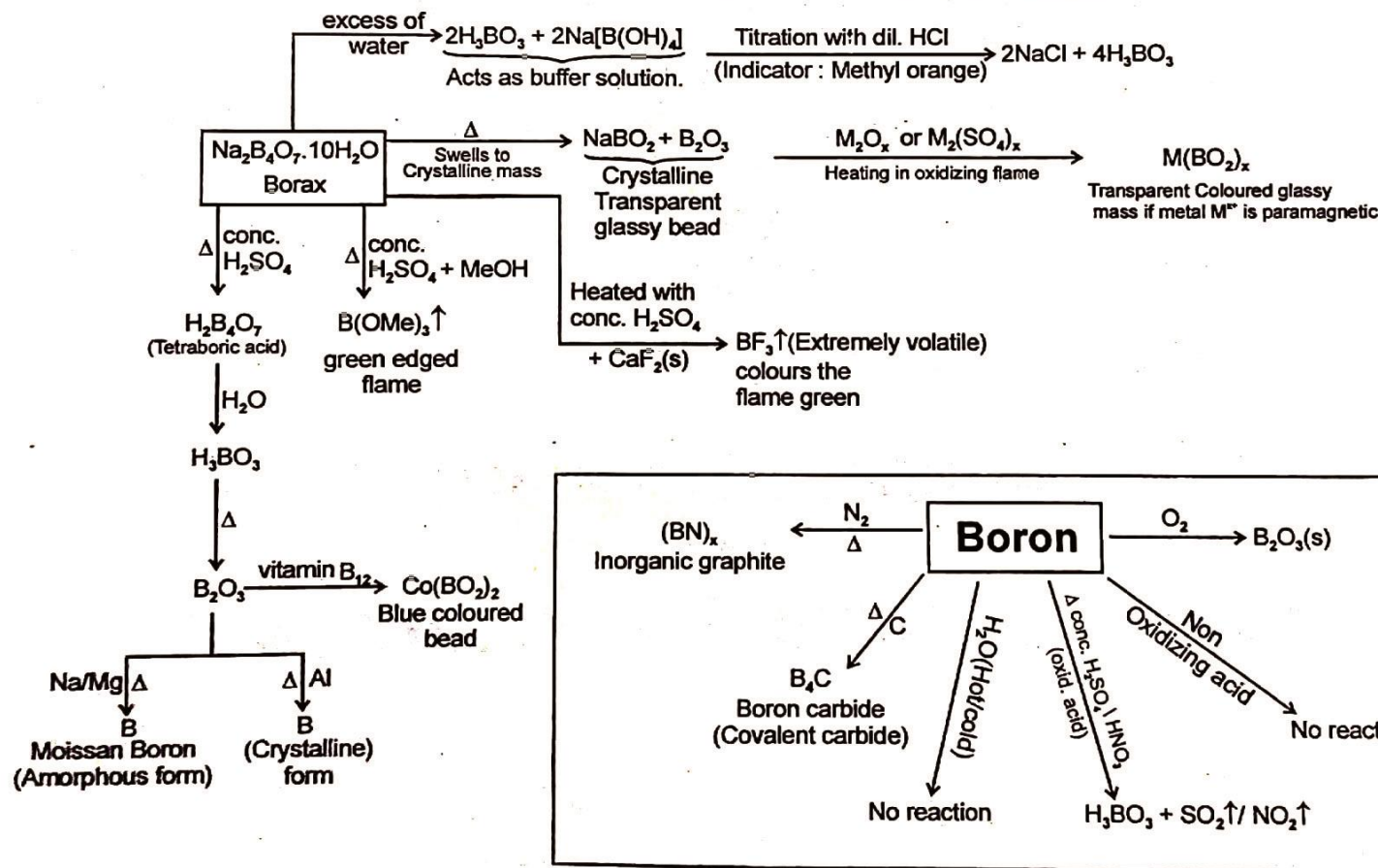


Borax $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$

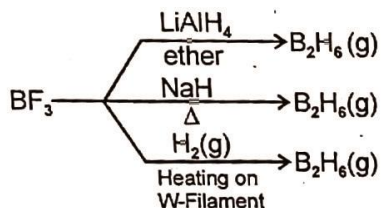


Note : $\Rightarrow$ On heating borax initially swells due to escape of water molecules and then it finally converts into crystalline mass consisting of NaBO_2 and B_2O_3 , similarly alums also swell on heating due to escape of water molecules but they convert into amorphous powder.
 $\Rightarrow$ Aqueous solution of borax is used as primary standard for titrating against acids. like HCl.

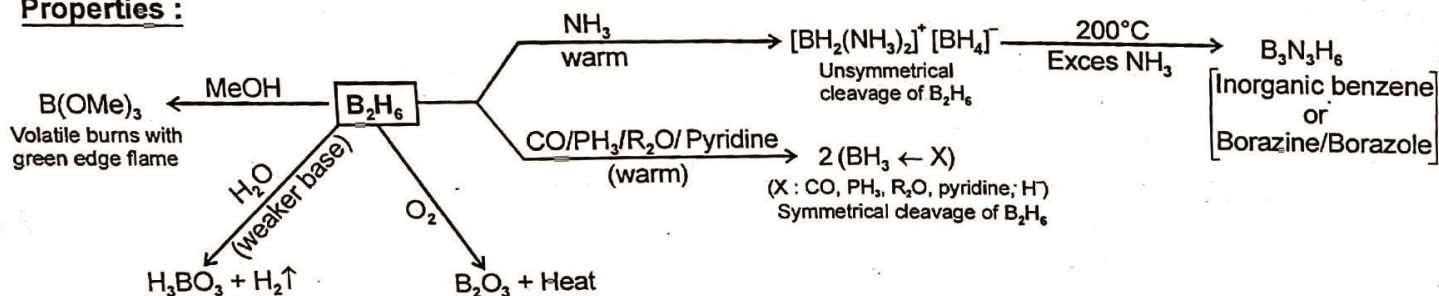
*

Diborane : B₂H₆

Preparation :

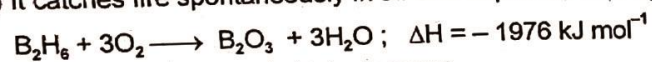


Properties :



Properties :

- (i) B₂H₆ is colourless gas and highly reactive (Boiling point 180 K)
- (ii) It catches fire spontaneously in air and explodes with O₂. Reaction with oxygen is extremely exothermic.



- (iii) Reaction with water is instantaneous.



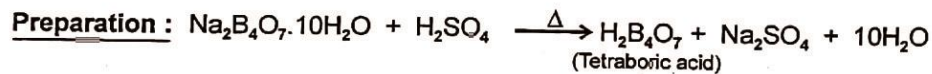
Symmetrical cleavage :



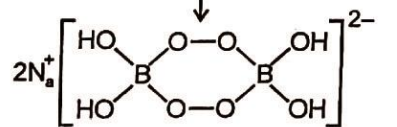
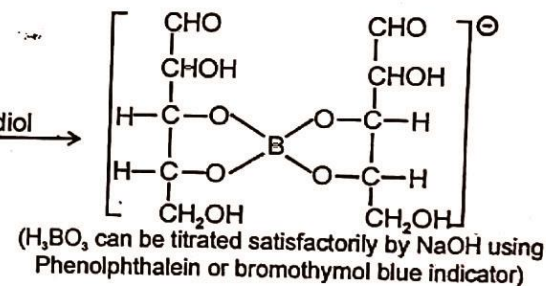
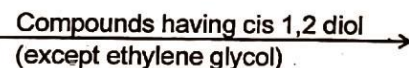
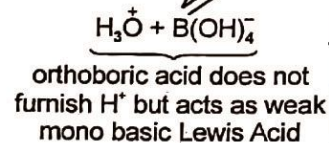
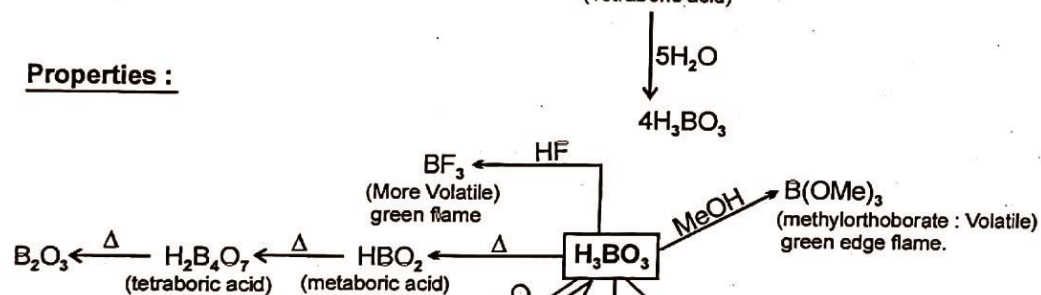
Symmetrical cleavage :



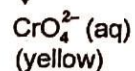
Orthoboric acid : H_3BO_3



Properties :

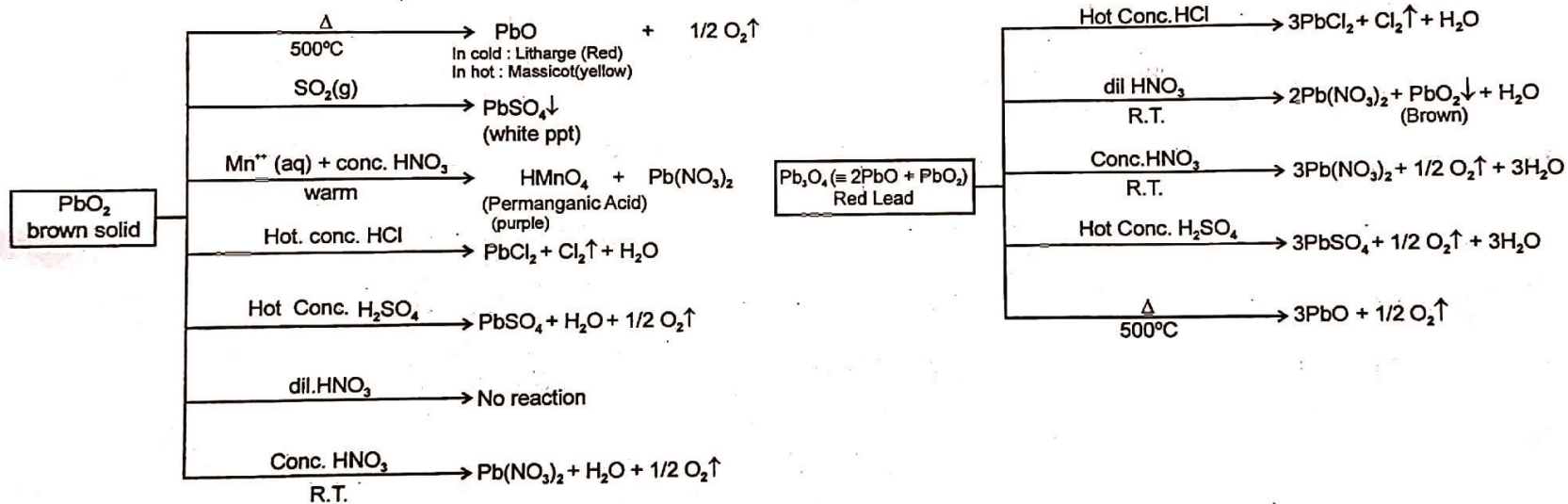


[Sodium peroxy borate.used
in washing powder as brightner]



Note : $\Rightarrow$ Solid boric acid is consisting of layered structure. Such layers are linked to one another by vander waals forces but within a layer H_3BO_3 molecules are bonded to one another by H-bonds.

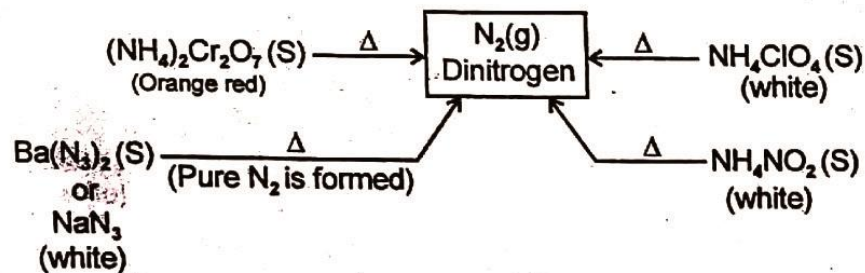
Oxides of Lead



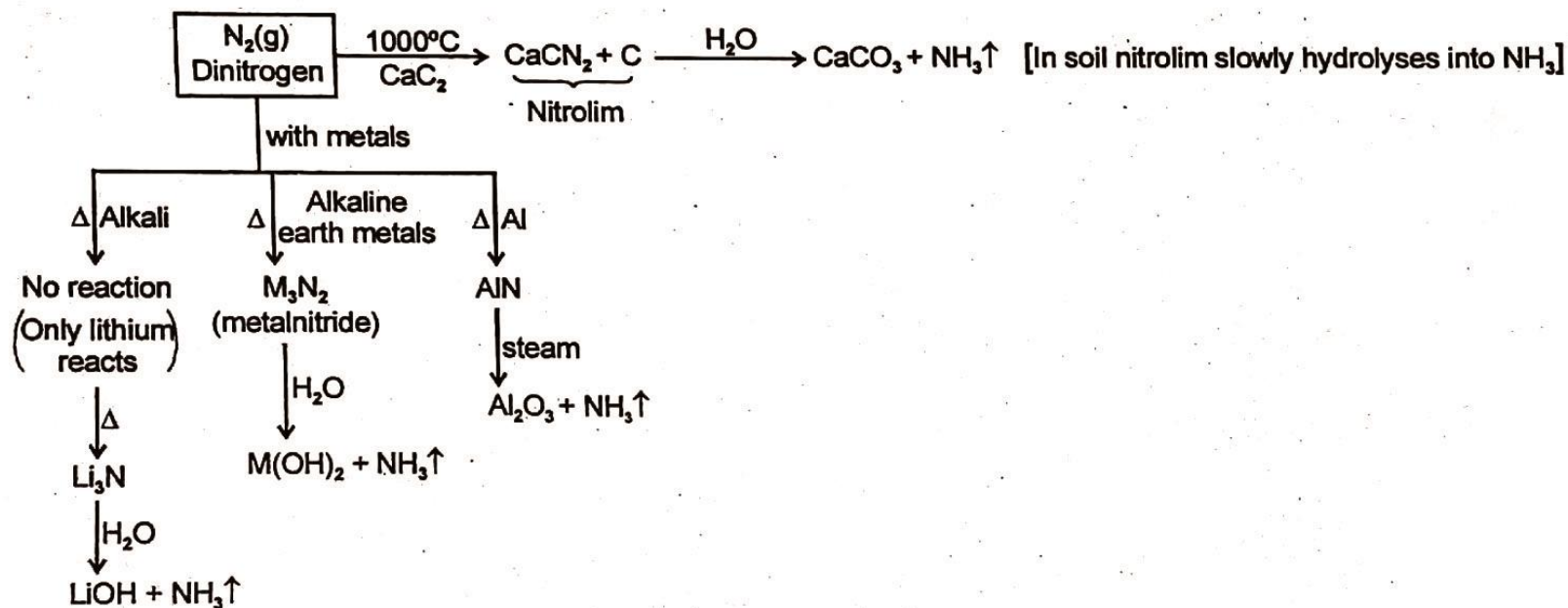
Note $\Rightarrow$ Both PbO and PbO₂ are amphoteric
 $\Rightarrow$ Compounds of Pb^{IV} are strong oxidizing agent.

Dinitrogen : N_2

Preparation :



Properties :



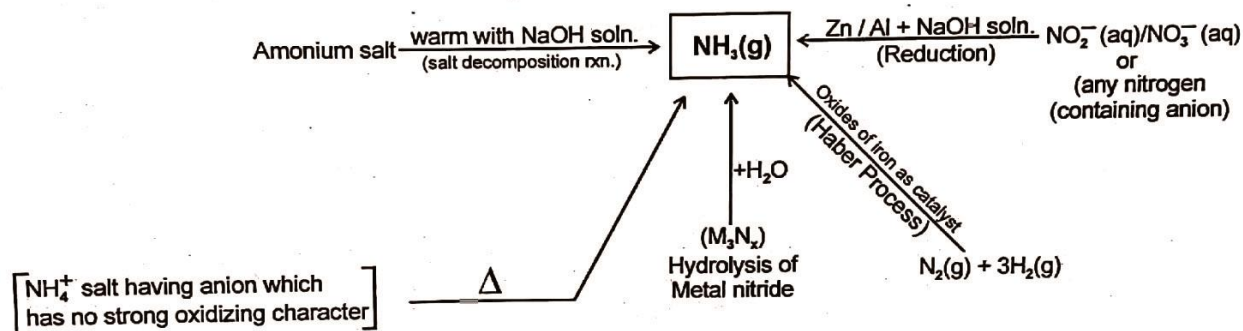
Note : $\Rightarrow$ Nitrogen gas can be obtained on heating those ammonium salts in which anionic part has strong oxidizing character.

$\Rightarrow$ Pure nitrogen is obtained on heating metal azides like $Ba(N_3)_2, NaN_3$

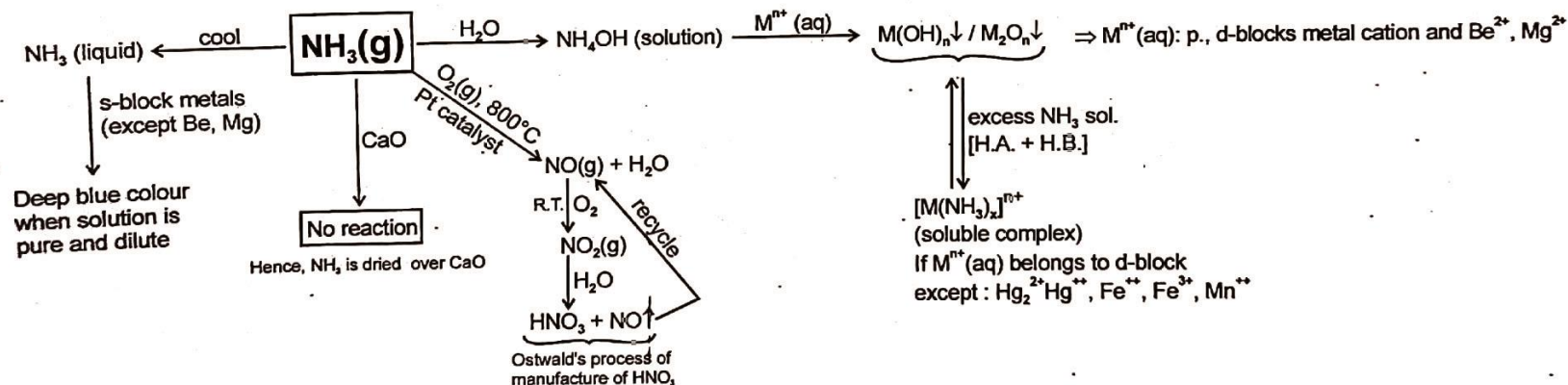
$\Rightarrow$ Nitrolim is used as fertilizer.

Ammonia : NH_3

Preparation

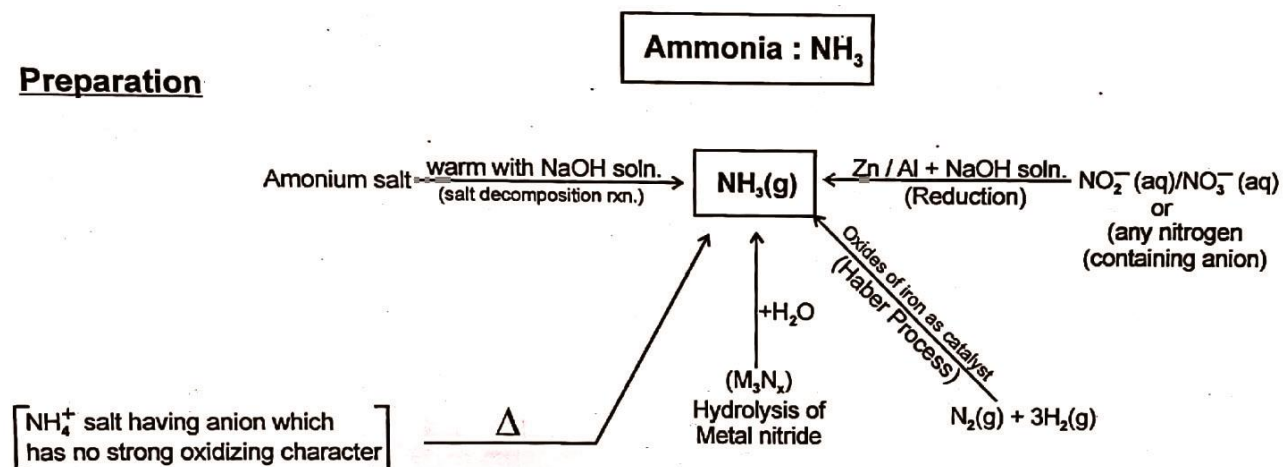


Properties :

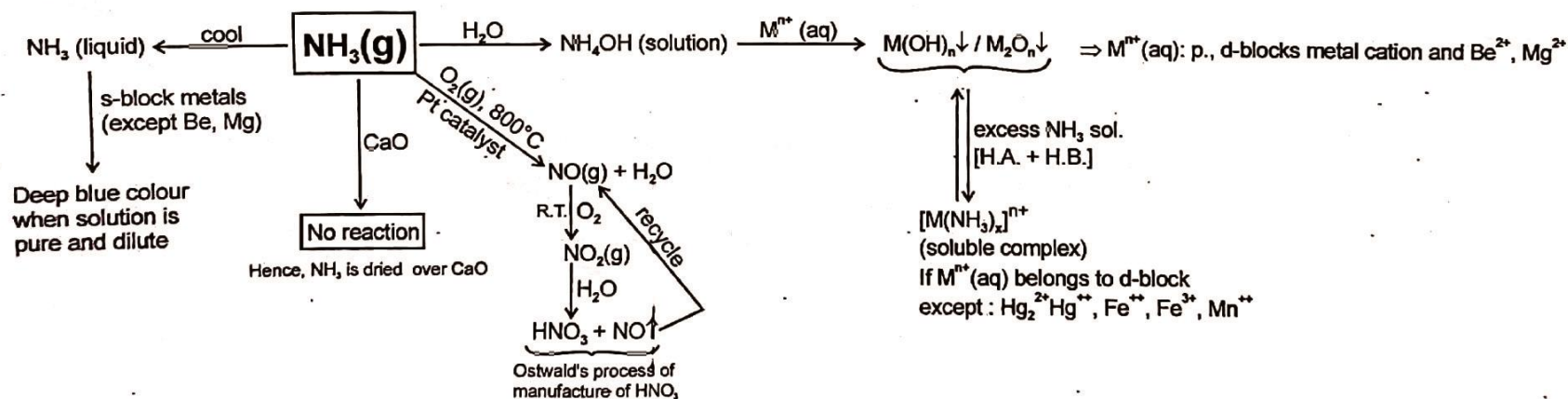


Note : $\Rightarrow \text{NH}_3$ gas is not dried by conc. H_2SO_4 , P_2O_5 , CaCl_2 .
 $\Rightarrow \text{NH}_3$ is dried by CaO .
 $\Rightarrow \text{Be}$ and Mg do not dissolve in liquid NH_3 due to high lattice and ionisation energies.

Preparation



Properties :

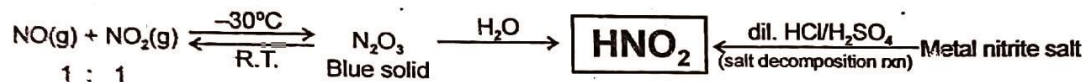


Note : $\Rightarrow \text{NH}_3$ gas is not dried by conc. H_2SO_4 , P_2O_5 , CaCl_2 .

$\Rightarrow \text{NH}_3$ is dried by **CaO**.

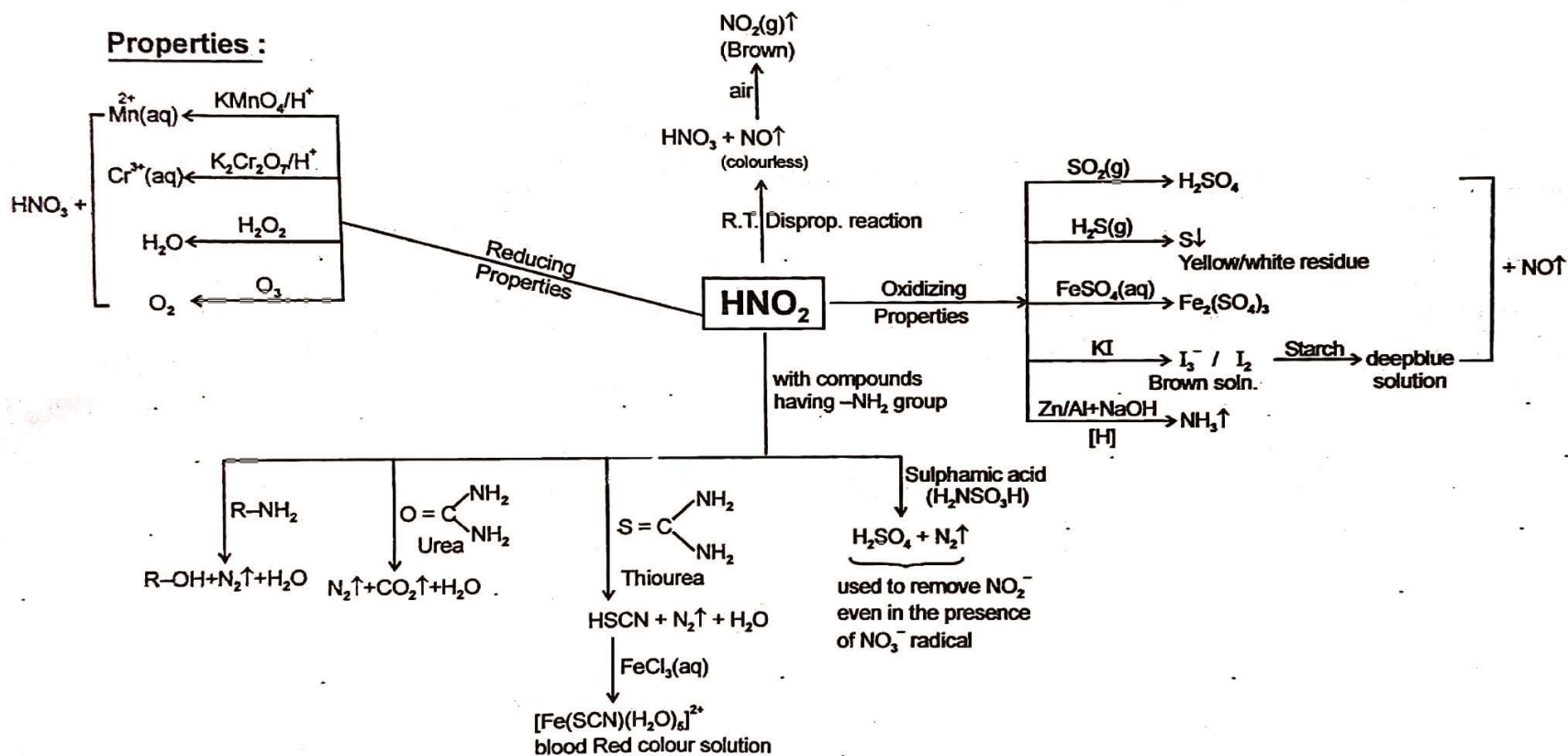
$\Rightarrow \text{Be}$ and **Mg** do not dissolve in liquid NH_3 due to high lattice and ionisation energies.

Preparation :



Nitrous Acid : HNO_2

Properties :



Note $\Rightarrow$ Free HNO_2 acid is not stored because it undergoes disproportionation into HNO_3 and NO at R.T.
 $\Rightarrow \text{N}_2\text{O}_3$ is the only oxide of nitrogen which is blue coloured.

Reactions of HNO₃ Acid

Oxidant (Different Conc. of HNO ₃)	Reductant (Metals / Non-metals / Compounds)	Main Products of Redox Reaction.
Very very dilute HNO ₃ (2%) (no oxidising character)	Mg, Mn	H ₂ ↑ + Metal nitrate
Very dilute HNO ₃ (6%) (weak oxidant)	Fe, Zn, Sn	NH ₄ NO ₃ + metalnitrate + H ₂ O
Dilute HNO ₃ (20%) (Mild oxidant)	Pb, Cu, Ag, Hg	NO↑ + metal nitrate + H ₂ O
	Fe, Zn	N ₂ O↑ + metalnitrate + H ₂ O
Conc. HNO ₃ (70%) (Strong oxidant)	Zn, Fe, Pb, Cu, Ag	NO ₂ ↑ + metal nitrate + H ₂ O
	Sn	NO ₂ ↑ + H ₂ SnO ₃ (metastannic acid)
	Au, Pt	No reaction } they dissolve in aqua regia due to chloro complex formation.
	Al, Cr, Co, Ni, Fe	Become Passive due to formation of oxide layer at their surfaces.
	S, C, P ₄ , Cl ₂ , Br ₂ , I ₂ (Non-metals)	NO ₂ ↑ + ic-form of respective oxy acid.
	As, Sb, Sn (Metalloids)	
Dil. HNO ₃ (20%) at R.T.	SO ₂ / H ₂ S / KI / FeS	<u>H₂SO₄ / S / I₂ / Fe₂(SO₄)₃</u> + NO↑ Respective products
Conc. HNO ₃ (70%)		<u>H₂SO₄ / H₂SO₄ / HIO₃ / Fe₂(SO₄)₃</u> + NO ₂ Respective products

Note ⇒ Behaviour of lead towards HNO₃ is same as those of Cu, Ag, Hg

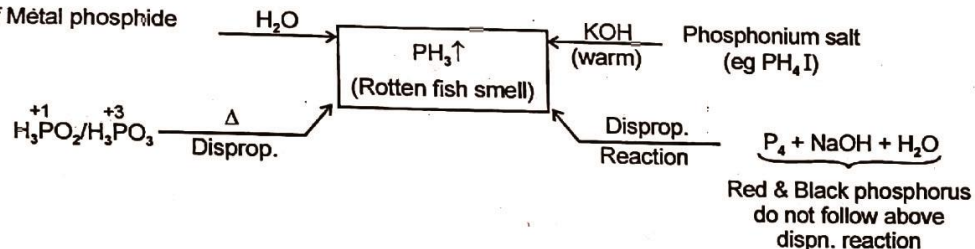
⇒ Pb, Cu, Ag, Hg do not react with very dilute HNO₃.

⇒ Ag does not dissolve in aqua regia but Au and Pt dissolve in aqua regia due to chloro complex formation.

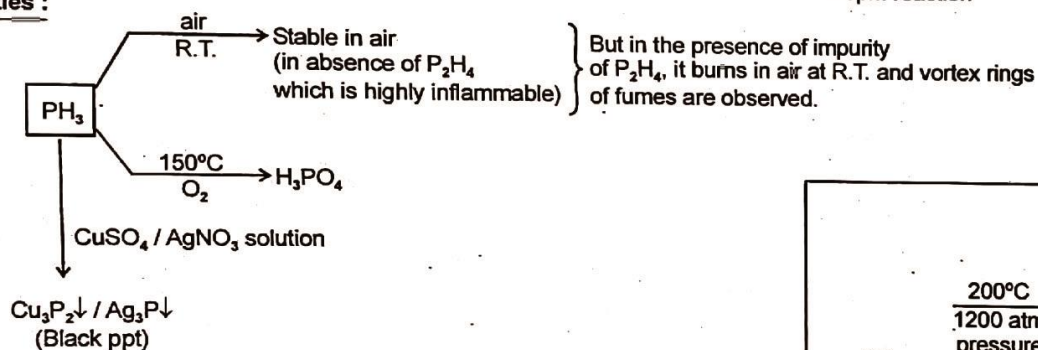
Phospine : PH_3

Preparation :

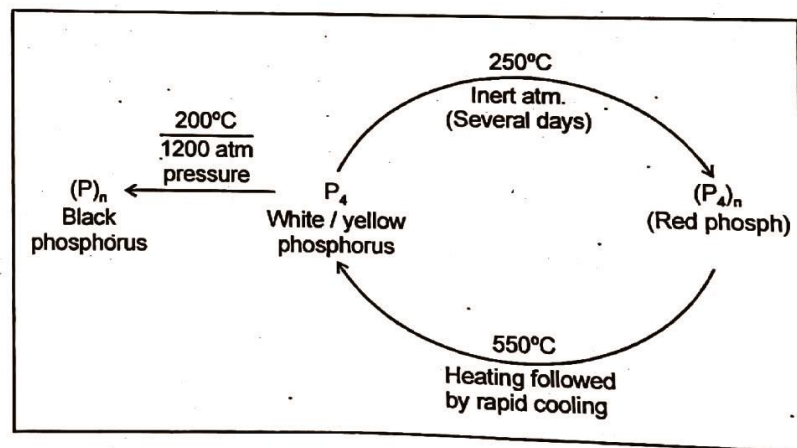
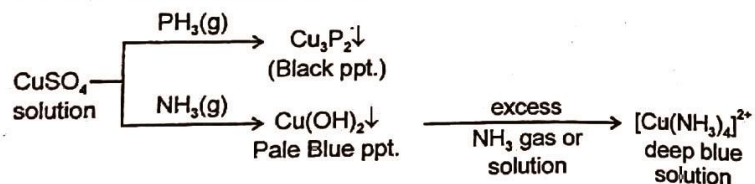
Hydrolysis of Metal phosphide



Properties :



Distinction between NH_3 and PH_3



Note ⇒ PH_3 forms metal phosphides while NH_3 forms metal hydroxides with water soluble metal salts.

⇒ Holme signal = mixture of $\text{Ca}_3\text{P}_2 + \text{CaC}_2$ (on hydrolysis gives PH_3 & C_2H_2 respectively which gives a signal to ships during night)